TEMPORAL GNN STREAM

ONTOLOGY TBOX STREAM

Supreme Court Narrative

Authentic Civil Disputes (2018–2026)

Case Facts (BLDS 2015 Art. 328):

"Buyer A deposited 500M VND to Seller B. Closing deadline passed. Dispute filed

Δt = 1,254 days later without curing."

Dynamic Multi-Relational Graph G(t)



Operative Latency: Δt = t_filing - t_breach

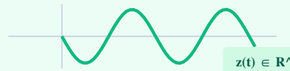
G(t), Δt

Multi-Feature Encoders

Continuous Bochner Kernel & PhoBERT

Bochner Spectral Time Kernel

$\Phi_d(\Delta t) = \sqrt{2/d}\,[\cos(\omega_k \Delta t), \sin(\omega_k \Delta t)]$



$z(t) \in \mathbb{R}^{64}$

Legal Domain Encoder (PhoBERT)

Statutory Clause & Relational Context

Relation Embed $e_r \in \mathbb{R}^{128}$

Litigant Covenants & Roles

Party Attributes x_u, x_v (768d)

Patch Vector $Z_t = [z(t) \| e_r \| x_v]$

Z_t

$H(t) \in \mathbb{R}^{N \times 128}$

Axiom-TGAT Layer × L

Statutory FFN

Gated State Projection W_s

Layer Normalization

Temporal Graph Attention

$\alpha_{uv}(t) = Softmax(Q_u (K_v + z(\Delta t))/\sqrt{d})$

Layer Normalization

Normative Arbiter (NH-DA)

Lex Specialis & Precedence DAG

Statutory Hierarchy Precedence:

- *Land Law 2024 > BLDS 2015 (Art 156)*
- *Usury Cap ceiling: r ≤ 20%/yr (Art 468)*

Neuro-Symbolic Gated Fusion:

$h_G^* = h_G \odot (1 + \sigma(W_g\, h_G) \cdot w_{domain})$

Enforces Lex Specialis in Latent Space

Axiomatic TBox Regularizer:

$L_{total} = L_{task} + \lambda_{onto} L_{onto} + \lambda_{state} L_{state}$

HermiT Certified Compliance: 99.12% LCR

Dung Defeasible Head & Adjudication

Abstract Argumentation Framework AF = ⟨A, R_defeat⟩ (Dung 1995)

A_bar (Procedural)

Lapse Δt > 1095d (Art 429)

STATUS: ACCEPTED

def

A_claim (Substantive)

Double Deposit Penalty (Art 328)

STATUS: DEFEATED

GE(AF) = lfp(F_AF) = { A_bar }

Unique Minimal Fixed Point: Procedural Bar overrules Substantive Claim

FINAL JUDICIAL VERDICT (y*)

✓ DISMISS: STATUTE OF LIMITATIONS EXPIRED (Art. 429)

Hits@1: 86.41 ± 0.38%

Latency: 1.5 ms (A30 GPU)

VCLO-Core OWL 2 DL Formal Ontology & HermiT Reasoner

35 Classes, 233 SROIQ Axioms verified with zero inconsistency (owl:Nothing = ∅)

TBox Coded Statutory Axioms

Axiom 1: Mutual Disjointness

ValidContract ⊓ VoidAbInitio ⊑ ⊥

Axiom 2: Non-Monotonic Curing (Art 129)

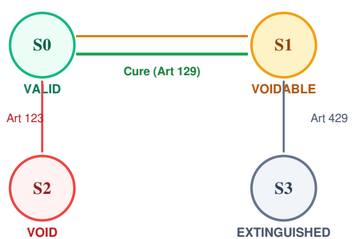
FormalDefect ⊓ (≥ 0.67 performed) ⊑ ¬VoidAbInitio ⊓ ValidContract

Axiom 3: Procedural Bar (Art 429)

Dispute ⊓ (Δt > 1095d) ⊓ ¬Tolling ⊑ BarredClaim ⊓ ExtinguishedState

HermiT Hypertableau: 0 Unsatisfiable

Markovian State Automaton M_state



Automaton Gating: $s_t \in \Delta^3$

Axiomatic Prior s_t

Backward Gradient Feedback: ∇ L_onto + ∇ L_state (TBox Margin Consistency)